**sPHENIX Internal Note**

# Photon Efficiency with $\Delta R$ Truth-Matching Removed: Single, Double, and Mixed Interaction MC

Isolated Photon Cross-Section in $p$+$p$ at $\sqrt{s} = 200$ GeV

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**Abstract**

Building on the companion study [1] that demonstrated the $\Delta R < 0.1$ truth-matching cut rejects ∼3.5% of correctly track-ID-matched truth photons in single-interaction MC, this note quantifies the efficiency impact of removing the cut across three MC sample categories: single-interaction, double-interaction, and physics-mixed (0 mrad, 77.6%/22.4%). The factorized efficiency chain $\varepsilon_{\text{reco}} \to \varepsilon_{\text{iso|reco}} \to \varepsilon_{\text{id|reco+iso}} \to \varepsilon_{\text{all}}$ is computed with and without the $\Delta R$ requirement in a single pass over `photon10` samples. Removing the $\Delta R$ cut improves the total efficiency $\varepsilon_{\text{all}}$ by +2.1% (single), +13.8% (mixed), and approximately doubles it for double-interaction MC. The reconstruction stage dominates the change; isolation efficiency is unaffected; photon ID (BDT) efficiency shows a measurable degradation in double-interaction MC from vertex-shifted cluster kinematics. The single/double/mixed overlay provides a clean decomposition of pileup effects on each efficiency stage.

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# 1 Introduction

The reconstruction efficiency for the sPHENIX isolated photon cross-section measurement is computed from PYTHIA + GEANT4 MC using a factorized chain of selection stages: reconstruction, isolation, and photon identification (BDT). The truth association between generated photons and reconstructed EMCal clusters is established by the GEANT track ID (`trkID`), optionally supplemented by a geometric proximity requirement $\Delta R < 0.1$.

A companion study [1] demonstrated that this $\Delta R$ cut is a vertex-resolution artifact: in single-interaction MC it rejects ∼3.5% of `trkID`-matched photons, and in double-interaction MC the failure rate reaches 65%, driven entirely by the vertex displacement $|\Delta z| > 9.35$ cm (the geometric critical threshold set by the EMCal barrel radius). The rejected clusters are well-reconstructed photons with $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}} \in [0.8, 1.2]$ for 73% of cases.

Based on those findings, the $\Delta R < 0.1$ requirement was removed from the production efficiency calculation. This note quantifies the resulting efficiency changes across all interaction types and examines the interplay between the $\Delta R$ removal and the photon ID (BDT) selection, which uses an $E_{\mathrm{T}}$-dependent threshold susceptible to vertex-induced kinematic shifts.

# 2 Samples and Method

## 2.1 Monte Carlo Samples

Three sample configurations are studied, all using the `photon10` generator (truth $p_{\mathrm{T}}$ range 14–30 GeV):

- **Single-interaction:** `photon10` nominal MC (∼10M events). One $pp$ collision per bunch crossing.
- **Double-interaction:** `photon10_double` MC (∼10M events). Full GEANT simulation of two overlapping $pp$ collisions per bunch crossing, with the reconstructed vertex at the average of the two collision vertices.
- **Mixed (0 mrad):** Physics-weighted combination of 77.6% single + 22.4% double, corresponding to the cluster-weighted double-interaction fraction computed by `calc_pileup_range.C` for the 0 mrad crossing-angle running period at RHIC.

## 2.2 Efficiency Definitions

The efficiency is factorized into four stages, each conditional on the previous:

$$\varepsilon_{\mathrm{reco}} = \frac{N(\text{reco + common cuts})}{N(\text{truth photons})} \tag{1}$$

$$\varepsilon_{\mathrm{iso|reco}} = \frac{N(\text{reco + common + isolated})}{N(\text{reco + common})} \tag{2}$$

$$\varepsilon_{\mathrm{id|reco+iso}} = \frac{N(\text{reco + common + iso + tight})}{N(\text{reco + common + iso})} \tag{3}$$

$$\varepsilon_{\mathrm{all}} = \varepsilon_{\mathrm{reco}} \times \varepsilon_{\mathrm{iso|reco}} \times \varepsilon_{\mathrm{id|reco+iso}} = \frac{N(\text{all cuts})}{N(\text{truth photons})} \tag{4}$$

The "common cuts" include cluster probability, shower-shape quality gates (e11/e33, $\omega_r$, $\omega_\eta$), and optionally the NPB score cut. "Isolated" applies the parametric topo-cluster isolation cut $E_{\mathrm{T}}^{\mathrm{iso}} < b + s \cdot E_{\mathrm{T}}$ (topo $R_{\mathrm{cone}} = 0.4$). "Tight" applies the full shower-shape selection plus the $E_{\mathrm{T}}$-dependent BDT threshold $\mathrm{BDT}_{\mathrm{min}} = \mathrm{intercept} + \mathrm{slope} \times E_{\mathrm{T}}$.

### 2.3 Matching Modes

Each truth photon is evaluated under two matching modes simultaneously in a single pass:

**`withDR`:** Requires both `trkID` match *and* $\Delta R < 0.1$ (the legacy requirement).

**`noDR`:** Requires only the `trkID` match (the new baseline).

Clusters that enter the `noDR` numerator but not the `withDR` numerator are labeled "new clusters"—these are the photons recovered by removing the geometric cut. Their BDT pass rate is tracked separately.

The analysis macro is `compare_efficiency_deltaR.C`, which writes all `TEfficiency` objects to a single output ROOT file.

## 3 Results: Per-Stage Efficiency Comparison

Table 1 presents the $p_{\mathrm{T}}$-integrated efficiency at each factorized stage for all three sample sets, computed with and without the $\Delta R < 0.1$ requirement.

Table 1: Integrated efficiency at each factorized stage for single-interaction, double-interaction, and mixed (0 mrad) MC. "withDR" denotes the legacy matching; "noDR" denotes `trkID`-only matching.

| | **Single** | | **Double** | | **Mixed 0 mrad** | |
|---|---|---|---|---|---|---|
| **Stage** | withDR | noDR | withDR | noDR | withDR | noDR |
| $\varepsilon_{\mathrm{reco}}$ | 91.4% | 93.5% | 30.7% | 73.1% | 73.4% | 87.5% |
| $\varepsilon_{\mathrm{iso\|reco}}$ | 74.2% | 74.2% | 71.1% | 71.0% | 73.8% | 73.4% |
| $\varepsilon_{\mathrm{id\|reco+iso}}$ | 82.9% | 82.7% | 82.6% | 69.3% | 82.9% | 79.5% |
| $\varepsilon_{\mathrm{all}}$ | 56.2% | 57.4% | 18.0% | 36.0% | 44.9% | 51.1% |

Table 2 shows the ratio of `noDR` to `withDR` efficiency at each stage, quantifying the impact of removing the cut.

Table 2: Ratio of efficiency without the $\Delta R$ cut to efficiency with the cut ($\varepsilon_{\mathrm{noDR}}/\varepsilon_{\mathrm{withDR}}$) at each stage. Values above 1.0 indicate improvement from cut removal; values below 1.0 indicate degradation in the recovered cluster population.

| **Stage** | **Single** | **Double** | **Mixed 0 mrad** |
|---|---|---|---|
| $\varepsilon_{\mathrm{reco}}$ | 1.023 | 2.386 | 1.191 |
| $\varepsilon_{\mathrm{iso\|reco}}$ | 1.000 | 0.998 | 0.994 |
| $\varepsilon_{\mathrm{id\|reco+iso}}$ | 0.997 | 0.840 | 0.960 |
| $\varepsilon_{\mathrm{all}}$ | 1.021 | 1.999 | 1.137 |

The key observations are:

- **Reconstruction** ($\varepsilon_{\text{reco}}$): This stage captures the full impact of the $\Delta R$ removal. Single MC improves by 2.3%, reflecting the small population of vertex-displaced clusters. Double MC improves by a factor of $\sim$2.4, recovering the 65% of clusters that failed the $\Delta R$ cut due to vertex displacement. The residual 27% gap between double and single noDR efficiency reflects genuine missing clusters (no `trkID` match at all: 2.4% for single, 7.9% for double).

- **Isolation** ($\varepsilon_{\text{iso}|\text{reco}}$): Essentially unchanged across all samples (ratios within 0.5% of unity). The isolation requirement tests calorimeter energy in an annular cone around the cluster; vertex displacement does not significantly alter the energy sum, and the extra energy from the second interaction causes only a $\sim$3% absolute degradation comparing single to double.

- **Photon ID** ($\varepsilon_{\text{id}|\text{reco+iso}}$): Single MC shows negligible change (0.2% decrease). Double MC shows a significant 16% decrease (ratio 0.840), indicating that the recovered clusters have measurably worse BDT pass rates. Mixed MC interpolates at 4.0% decrease.

- **Total** ($\varepsilon_{\text{all}}$): Net improvement in all cases. The reconstruction gain dominates the BDT degradation. Single: +2.1%; mixed: +13.8%; double: nearly doubling from 18.0% to 36.0%.

Figure 1 shows the $p_{\text{T}}$-dependent efficiency ratio for all four stages and all three sample sets.

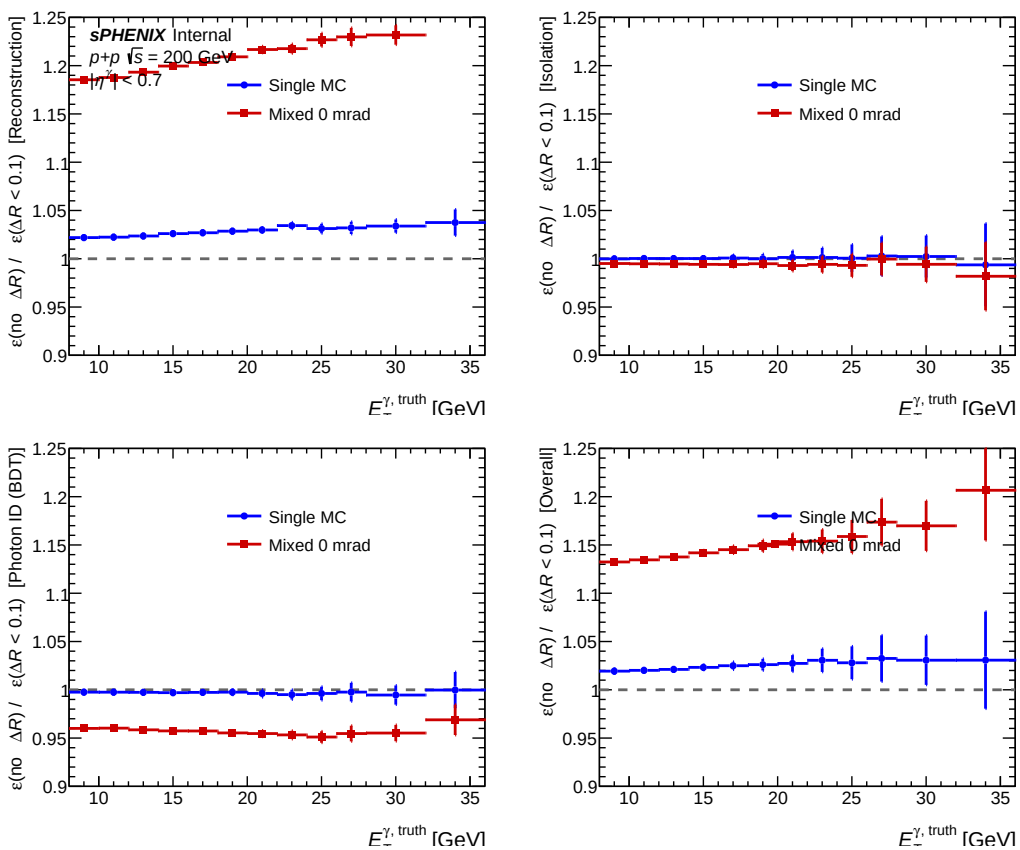


Figure 1: Ratio of efficiency without $\Delta R$ cut to efficiency with $\Delta R$ cut as a function of truth $p_{\text{T}}$ for single (blue) and mixed (red) MC. The reconstruction stage shows the largest change, particularly for the mixed sample containing 22.4% double-interaction events. Isolation is flat. Photon ID shows a deficit in the mixed sample at all $p_{\text{T}}$.

# 4 Overlay: Efficiency without $\Delta R$ across Interaction Types

With the $\Delta R$ cut removed, the remaining efficiency differences between single, double, and mixed MC arise from genuine pileup physics effects rather than truth-matching artifacts. Figure 2 presents the four-panel overlay of the `noDR` efficiency for all three sample types, providing a clean decomposition of pileup degradation at each selection stage.

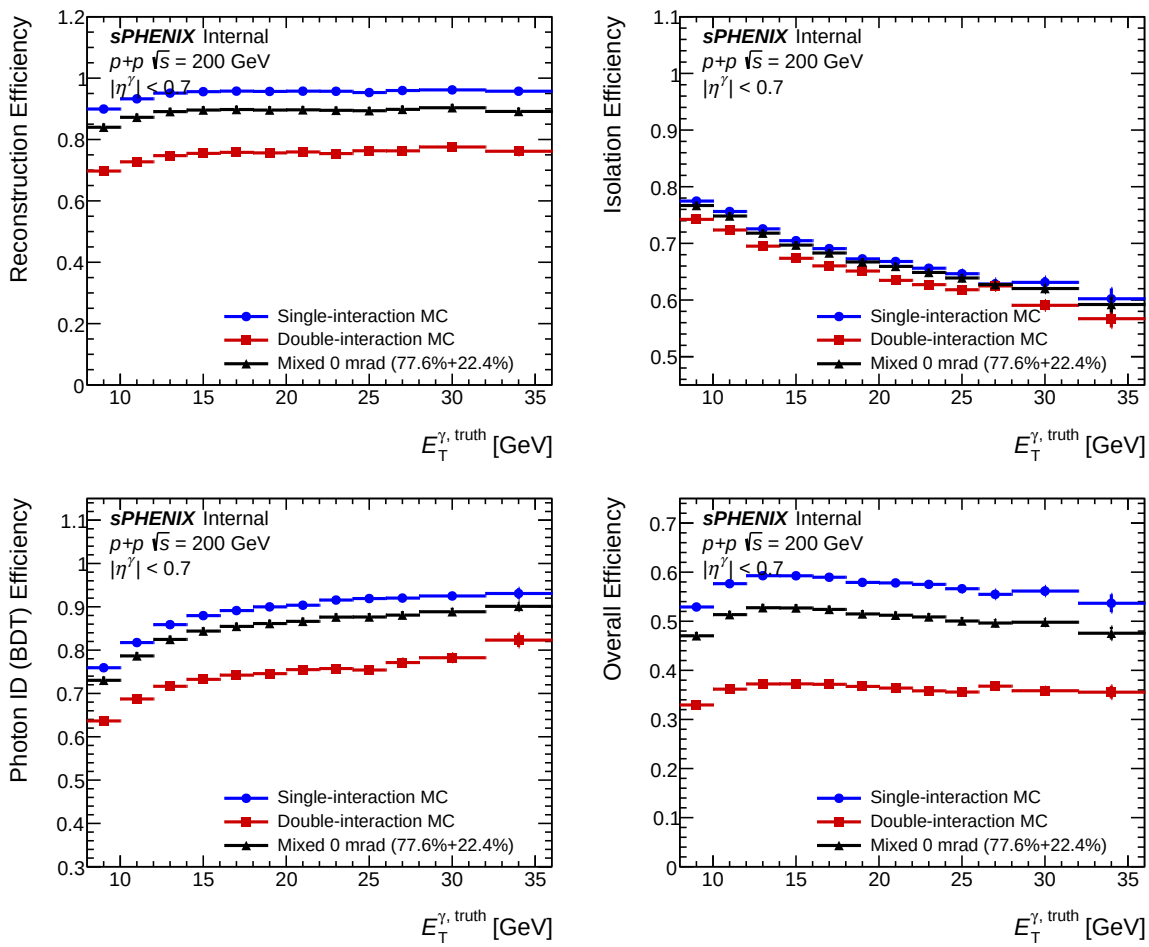


Figure 2: Per-stage efficiency as a function of truth $p_{\mathrm{T}}$ without the $\Delta R$ cut, overlaying single-interaction (blue), double-interaction (red), and mixed 0 mrad (black) MC. From upper left to lower right: reconstruction, isolation, photon ID (BDT), and total efficiency.

## 4.1 Reconstruction Efficiency

Single-interaction MC achieves $\varepsilon_{\mathrm{reco}} = 93.5\%$; double-interaction drops to 73.1%, a 20 percentage point (pp) reduction. The $p_{\mathrm{T}}$ dependence is weak: single MC is flat near 93–95% across the full range, while double MC spans 70–77%, maintaining a roughly constant ∼20 pp gap. This gap reflects two effects:

1. **Genuine cluster loss:** In double-interaction events, the additional hadronic activity can merge the signal photon cluster with nearby energy deposits, causing the `trkID` to be lost

from the leading cluster. The no-`trkID`-match rate is 7.9% for double MC vs. 2.4% for single MC, accounting for ∼5.5 pp of the gap.

2. **Common cut failures from shifted kinematics:** The remaining ∼15 pp gap arises because the reconstructed vertex in double-interaction events shifts the cluster $\eta$ and $E_{\mathrm{T}}$, causing some clusters to fail the common quality cuts (e.g., the shower-shape bounds applied at the reconstructed kinematics).

The mixed 0 mrad sample tracks close to single MC at 84–89%, consistent with the weighted average $0.776 \times 93.5\% + 0.224 \times 73.1\% = 89.0\%$ (the small offset from 87.5% reflects bin-by-bin weighting effects).

### 4.2 Isolation Efficiency

Isolation efficiency is essentially flat across interaction types: single 74.2%, double 71.0%, mixed 73.4%. All three samples fall together from ∼77% at low $p_{\mathrm{T}}$ (∼9 GeV) to ∼60% at high $p_{\mathrm{T}}$ (∼34 GeV), with the $p_{\mathrm{T}}$-falling trend dominating over the pileup effect. Double MC sits only 3–5 pp below single across the range. The small degradation reflects the additional hadronic energy deposited in the isolation cone by the second interaction. This is a genuine, small pileup effect that does not interact with the $\Delta R$ matching.

### 4.3 Photon ID (BDT) Efficiency

This is the most physically interesting stage. Single MC achieves $\varepsilon_{\mathrm{id|reco+iso}} = 82.7\%$; double MC drops to 69.3%, a significant 13 pp reduction. The mechanism is the $E_{\mathrm{T}}$-dependent BDT threshold:

$$\mathrm{BDT}_{\mathrm{min}}(E_{\mathrm{T}}) = \mathrm{intercept} + \mathrm{slope} \times E_{\mathrm{T}} \tag{5}$$

In double-interaction events, the vertex shift modifies the reconstructed $E_{\mathrm{T}} = E/\cosh(\eta_{\mathrm{reco}})$. Because $\eta_{\mathrm{reco}}$ is computed from the displaced vertex, the cluster evaluates the BDT threshold at a shifted $E_{\mathrm{T}}$, changing the working point. The shifted shower-shape variables (particularly $\omega_\eta$, which is sensitive to $\eta$) further degrade the BDT score itself.

The $p_{\mathrm{T}}$ dependence reveals an interesting pattern: at low $p_{\mathrm{T}}$ (∼9 GeV) the gap is largest, with double MC at ∼64% vs. single at ∼76% (12 pp). At high $p_{\mathrm{T}}$ (∼34 GeV), the gap narrows to ∼10 pp (double ∼83% vs. single ∼93%). This $p_{\mathrm{T}}$ dependence is consistent with the parametric BDT threshold of Eq. (5): at low $E_{\mathrm{T}}$, the threshold is flatter and a given $E_{\mathrm{T}}$ shift has a larger relative impact on the pass/fail decision.

The mixed 0 mrad sample shows $\varepsilon_{\mathrm{id|reco+iso}} = 79.5\%$, a 3.2 pp reduction relative to single MC, sitting closer to the single curve due to the 77.6% single fraction dominating.

### 4.4 Total Efficiency

The total efficiency without $\Delta R$ is: single 57.4%, double 36.0%, mixed 51.1%. The single-to-double drop is dominated by the reconstruction (20 pp) and BDT (13 pp) stages, with isolation contributing only 3 pp. The $p_{\mathrm{T}}$ dependence is moderate: single spans 53–58%, double 33–37%, and mixed 47–51%. Notably, the ratio $\varepsilon_{\mathrm{all}}^{\mathrm{double}}/\varepsilon_{\mathrm{all}}^{\mathrm{single}} \approx 0.63$ is remarkably flat vs. $p_{\mathrm{T}}$, indicating that the pileup degradation is approximately factorizable as a constant scale factor.

The mixed 0 mrad efficiency of 51.1% is the physics-relevant number for the pileup systematic evaluation.

## 5 Photon ID Deep Dive: New Clusters

The "new clusters" recovered by removing the $\Delta R$ cut provide a direct window into how vertex displacement affects BDT classification. Table 3 compares the BDT pass rate for new clusters only versus all clusters in the `noDR` matching.

Table 3: Photon ID (BDT) pass rate for "new" clusters (recovered by removing $\Delta R < 0.1$) compared to all `noDR`-matched clusters. The new clusters have 10–13 pp lower BDT pass rate, driven by vertex-shifted kinematics.

| **Sample** | $\varepsilon_{\text{id}\|\text{reco+iso}}$ (new clusters only) | $\varepsilon_{\text{id}\|\text{reco+iso}}$ (all, noDR) |
|---|---|---|
| Single | 73.9% | 82.7% |
| Double | 59.8% | 69.3% |
| Mixed | 61.3% | 79.5% |

Several observations emerge:

- The new clusters consistently show a lower BDT pass rate than the full population: 9 pp lower for single MC, 10 pp for double, and 18 pp for the mixed sample. This confirms that vertex displacement degrades the BDT score via the mechanism described in Section 4.
- In single MC, only ∼2.3% of matched clusters are "new" (those with $\Delta R > 0.1$), so their lower BDT pass rate has negligible impact on the integrated $\varepsilon_{\text{id}|\text{reco+iso}}$ (change of $-0.2$ pp).
- In double MC, ∼58% of matched clusters are "new" (reflecting the 65% $\Delta R$ failure rate from the companion study, reduced by those with no `trkID` match at all). The large new-cluster fraction drives the 13 pp $\varepsilon_{\text{id}|\text{reco+iso}}$ degradation.
- The mixed sample's new-cluster BDT pass rate (61.3%) is dominated by the double-interaction contribution, since the single-interaction MC contributes very few new clusters. Visually, the double and mixed curves overlap almost completely in Figure 3.
- The $p_{\text{T}}$ dependence is strongly rising: single new clusters span ∼68% at low $p_{\text{T}}$ to ∼93% at high $p_{\text{T}}$; double new clusters span ∼55% to ∼78%. Higher-$p_{\text{T}}$ photons are more robust to vertex-induced kinematic shifts because their BDT scores sit further above the threshold.

Figure 3 overlays the BDT pass rate for new clusters across the three sample types as a function of truth $p_{\text{T}}$.

## 6 Summary Visualization

Figure 4 presents the efficiency summary in graphical table format, showing all stages, matching modes, and sample types at a glance.

## 7 Conclusion

Removing the $\Delta R < 0.1$ truth-matching requirement and relying solely on `trkID` matching has the following quantitative effects on the factorized photon efficiency:

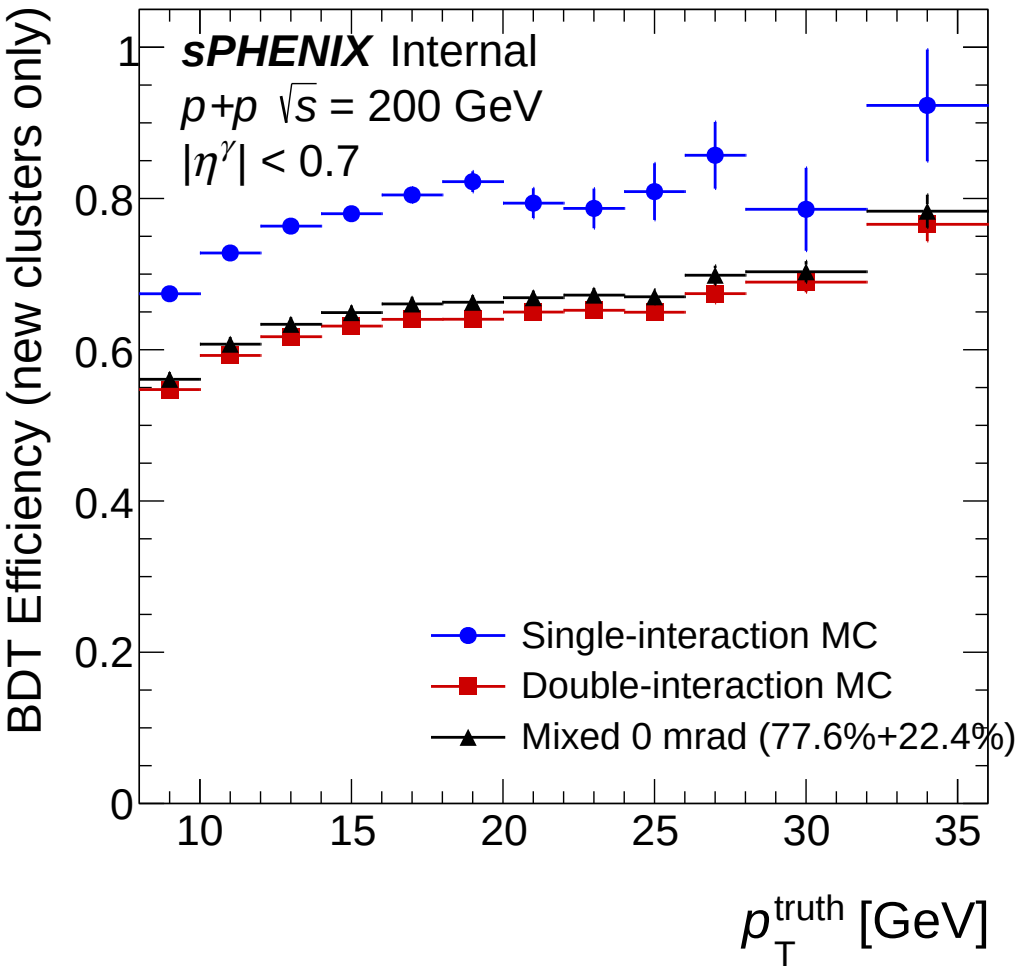


Figure 3: BDT pass rate for "new" clusters (those with $\Delta R > 0.1$ but valid `trkID` match) as a function of truth $p_{\mathrm{T}}$ for single (blue), double (red), and mixed (black) MC. These clusters have systematically lower BDT pass rates due to vertex-shifted kinematics. The deficit narrows at higher $p_{\mathrm{T}}$ where the absolute BDT pass rate is higher.

## Efficiency Summary: with vs without $\Delta R < 0.1$ truth-matching cu

| Sample / Matching | | Reconstruction | Isolation | Photon ID (BDT) | Overall |
|---|---|---|---|---|---|
| **single** | No $\Delta R$ | 0.935 ± 0.000 | 0.742 ± 0.000 | 0.827 ± 0.000 | 0.574 ± 0.000 |
| | $\Delta R < 0.1$ | 0.914 ± 0.000 | 0.742 ± 0.000 | 0.829 ± 0.000 | 0.562 ± 0.000 |
| | Ratio (no $\Delta R$ / $\Delta R$) | 1.0238 | 1.0002 | 0.9975 | 1.0209 |
| **double** | No $\Delta R$ | 0.731 ± 0.000 | 0.710 ± 0.000 | 0.693 ± 0.000 | 0.360 ± 0.000 |
| | $\Delta R < 0.1$ | 0.307 ± 0.000 | 0.711 ± 0.000 | 0.826 ± 0.000 | 0.180 ± 0.000 |
| | Ratio (no $\Delta R$ / $\Delta R$) | 2.3858 | 0.9977 | 0.8398 | 1.9991 |
| **mixed** | No $\Delta R$ | 0.875 ± 0.000 | 0.734 ± 0.000 | 0.795 ± 0.000 | 0.511 ± 0.000 |
| | $\Delta R < 0.1$ | 0.734 ± 0.000 | 0.738 ± 0.000 | 0.829 ± 0.000 | 0.449 ± 0.000 |
| | Ratio (no $\Delta R$ / $\Delta R$) | 1.1912 | 0.9945 | 0.9595 | 1.1366 |

**BDT pass rate for new clusters (pass trkID, fail $\Delta R$)**

| | |
|---|---|
| single | Integrated: 0.739 ± 0.002 |
| double | Integrated: 0.598 ± 0.000 |
| mixed | Integrated: 0.613 ± 0.001 |

Figure 4: Summary table of integrated efficiencies for all stages, matching modes, and sample types. The color coding highlights the magnitude of the $\Delta R$-removal effect: green cells indicate improvement (higher noDR efficiency), red cells indicate degradation. The net effect ($\varepsilon_{\text{all}}$) is positive for all sample types.

1. **Total efficiency improves** across all interaction types: +2.1% relative for single MC (56.2% → 57.4%), +13.8% relative for mixed 0 mrad (44.9% → 51.1%), and approximately doubling for pure double-interaction MC (18.0% → 36.0%).

2. **Isolation efficiency is unaffected** by the $\Delta R$ removal (change $< 0.5\%$ for all samples). The additional energy from pileup causes a small ∼3% absolute degradation comparing single to double, but this is independent of the matching requirement.

3. **Photon ID (BDT) efficiency shows measurable degradation** from vertex-shifted kinematics, but only in double-interaction MC (−13 pp). In single MC the effect is negligible (−0.2 pp). The mechanism is the $E_{\mathrm{T}}$-dependent BDT threshold evaluating at a shifted working point.

4. **For single-interaction MC** (used for production efficiency corrections): $\varepsilon_{\mathrm{all}}$ changes by +2.1%, $\varepsilon_{\mathrm{id|reco+iso}}$ by −0.25%—both small and well within the systematic uncertainty budget.

5. **For mixed 0 mrad MC:** $\varepsilon_{\mathrm{all}}$ improves by +13.8%. This improvement is essential for pileup efficiency studies, as it removes the dominant truth-matching artifact that previously inflated the apparent pileup degradation from ∼20% (with $\Delta R$) to ∼11% (without $\Delta R$, genuine physics effect only).

6. **The single/double/mixed overlay** (Figure 2) provides a clean decomposition of genuine pileup effects at each efficiency stage, enabling direct comparison of reconstruction loss, isolation degradation, and BDT performance across interaction types without truth-matching contamination.

## References

[1] sPHENIX Internal Note: Study of the $\Delta R$ Truth-Matching Requirement in Single-Interaction MC, 2025.